

Mihir Bhagatwala

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EDUCATION

University of California, Santa Cruz

B.S. in Computer Engineering

Santa Cruz, CA

Jun. 2027 (Expected)

– Cumulative GPA: 3.52

– Coursework: Signals & Systems, Circuit Theory, Systems Embedded Engineering, Logic Design (RTL Design & Validation), Data Structures & Algorithms, Computer Architecture, Electromagnetism, Thermodynamics, Probability, Statistics

The Bishop's School

High School Diploma

La Jolla, CA

Jun. 2023

– Cumulative GPA: 3.5

EXPERIENCE

Charging Board Engineer

Oct. 2025 - Present

Formula Slug

University of California, Santa Cruz

- Designed a compact **low-voltage PCB (KiCad)** for **SAE J1772** signaling, incorporating **TSMP-based power regulation** for efficient and isolated low-voltage rail generation.
- Implemented **CAN, SPI, and I²C interfaces** to transmit real-time charging status, voltage, and diagnostic data to the **onboard vehicle display** and vehicle control modules.
- Integrated voltage-sensing circuits and **safety interlocks**; validated power integrity and communication robustness via **bench testing, simulation**, and iterative hardware revisions for the 2026 FSAE competition.

Vision Engineer

Jan. 2025 - Present

Slugbotics

University of California, Santa Cruz

- Assembled MATE ROV, designed/fabricated waterproof **CAD** camera mounts, integrated hardware for **stable real-time video streaming**.
- Developed an **OpenCV** vision model (**image segmentation**) for shipwreck detection, using preprocessing to enhance **contour detection**.
- Integrated vision system with controls to support **semi-autonomous visual targeting** and operator assistance.

Research Assistant – Electronics Lab

Jan. 2025 – Present

Santa Cruz Institute for Particle Physics

Santa Cruz, CA

- Developed **optical instrumentation subsystems** for a LANL prototype, utilizing **38 kHz LED emitters** and **optical filters** to precisely simulate and verify isotope emission spectra.
- Designed high-fidelity **optical receiver circuits** to validate light emission frequency and signal integrity, ensuring compliance with strict experimental targets.
- Built **analog power-delivery circuits** with noise suppression and **over-voltage protection** to ensure stable operation of Arduino and Raspberry Pi controllers.
- Programmed **Arduino firmware** for precise 38 kHz pulse modulation and **Raspberry Pi scripts** to automate test cycles and synchronize timestamped data acquisition.

Teaching Assistant

Jun. 2023 - Aug. 2023

Art of Problem Solving (AoPS) Academy

San Diego, CA

- Delivered lessons on **geometry, algebra**, and **number theory** to classes of 15+ students, clarifying complex problem-solving concepts.

AWARDS

CruzHacks 2025 Justice Track Winner

Apr. 2025

Project: VoteSmart

University of California, Santa Cruz

- Engineered a **web app** in 36 hours that simplified election propositions by generating **personalized summaries** for voters.
- Designed and secured backend systems with **FastAPI** and **MongoDB**, incorporating **hashing techniques** to protect user data.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, MATLAB, Assembly

Software and Libraries: OpenCV, FastAPI, MongoDB, GitHub, AMD Vivado

Hardware and EDA: Verilog HDL, FPGA programming, KiCad, LTSpice, Oscilloscopes, Arduino, Raspberry Pi, Breadboard design, Soldering

CAD and Fabrication: SolidWorks, Onshape, Bambu Studio, PrusaSlicer